

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

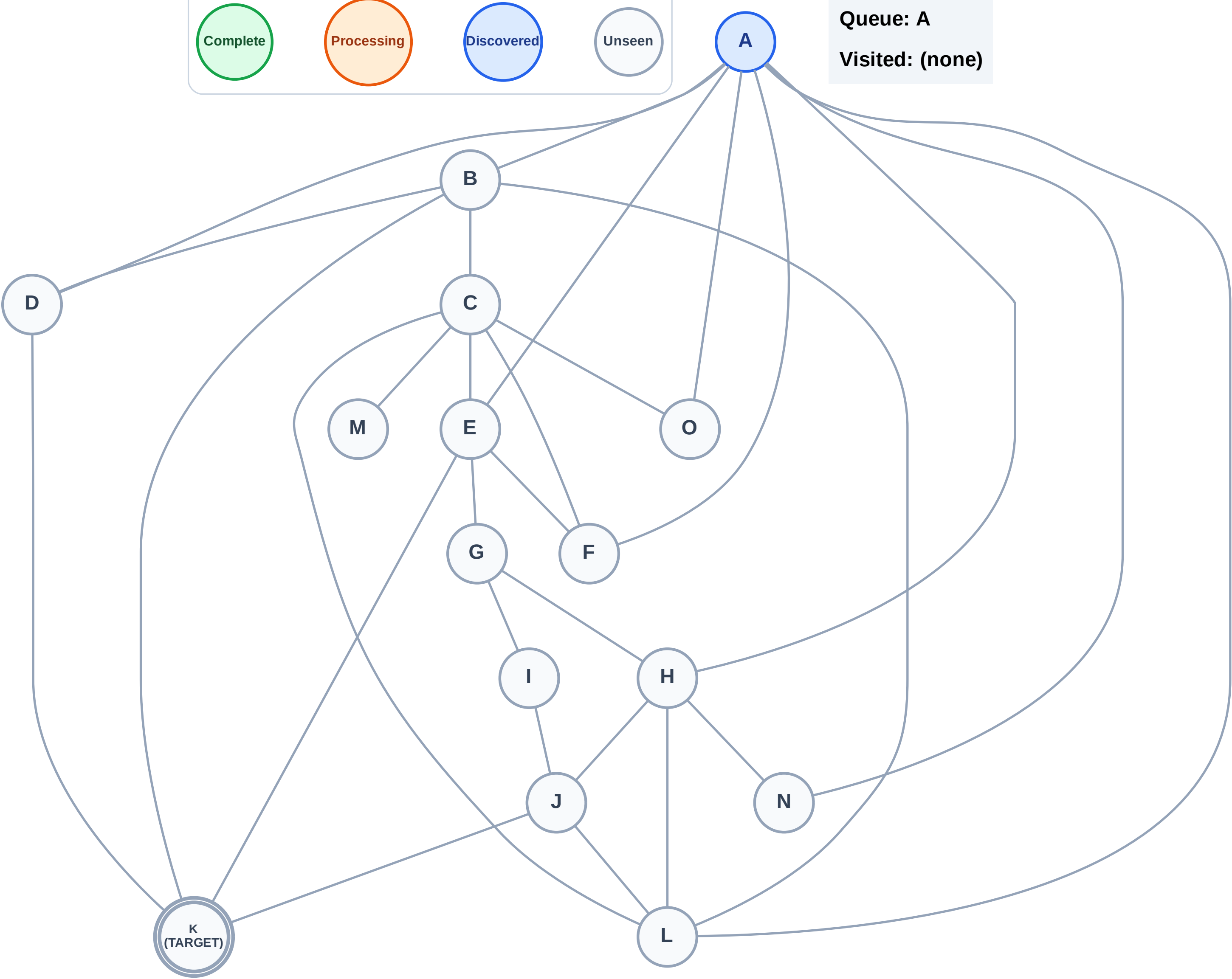
Processing

Discovered

Unseen

Queue: A

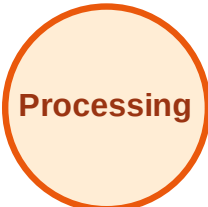
Visited: (none)



BFS Traversal

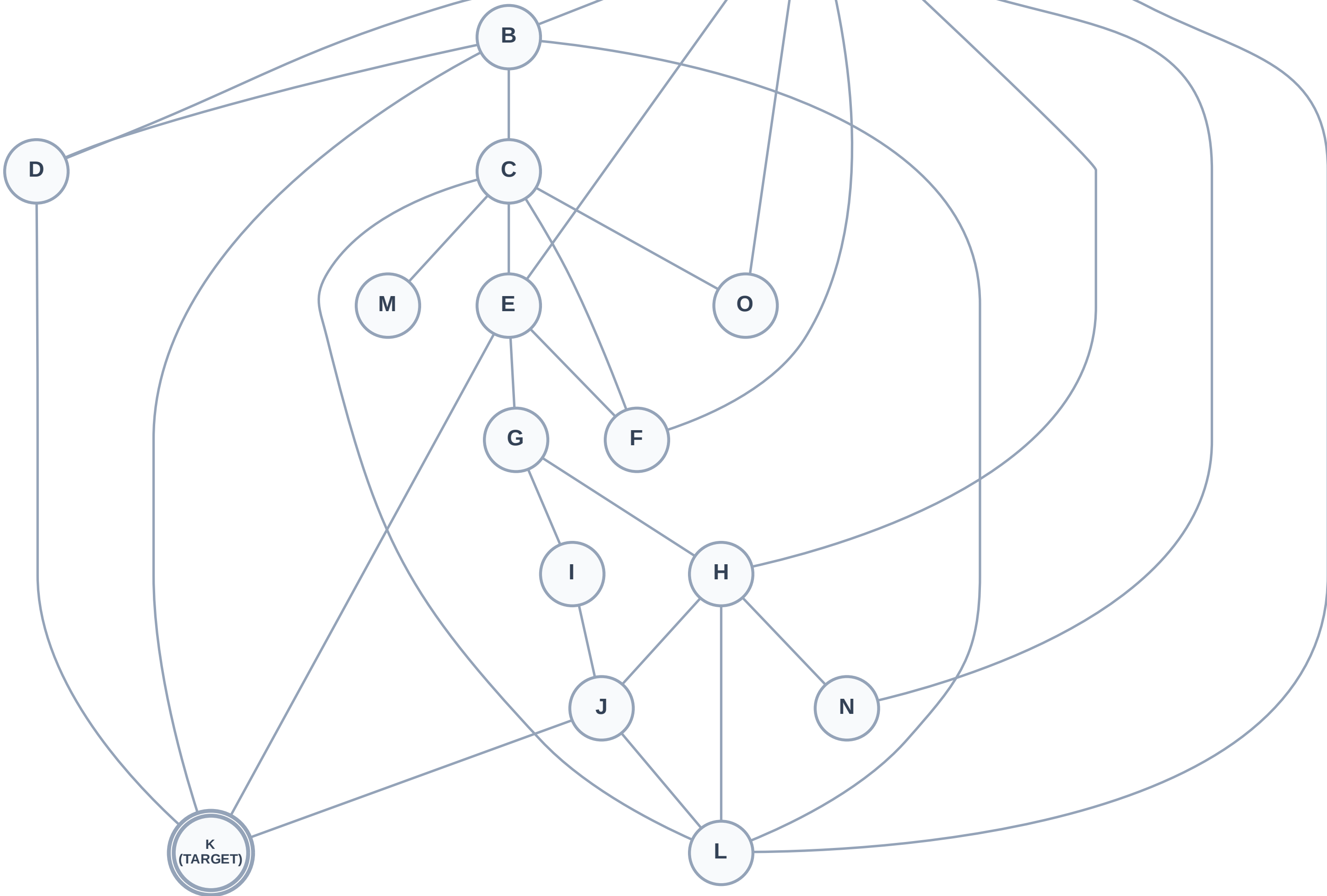
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, D, E, F, H, L, N, O from A

Legend

Complete

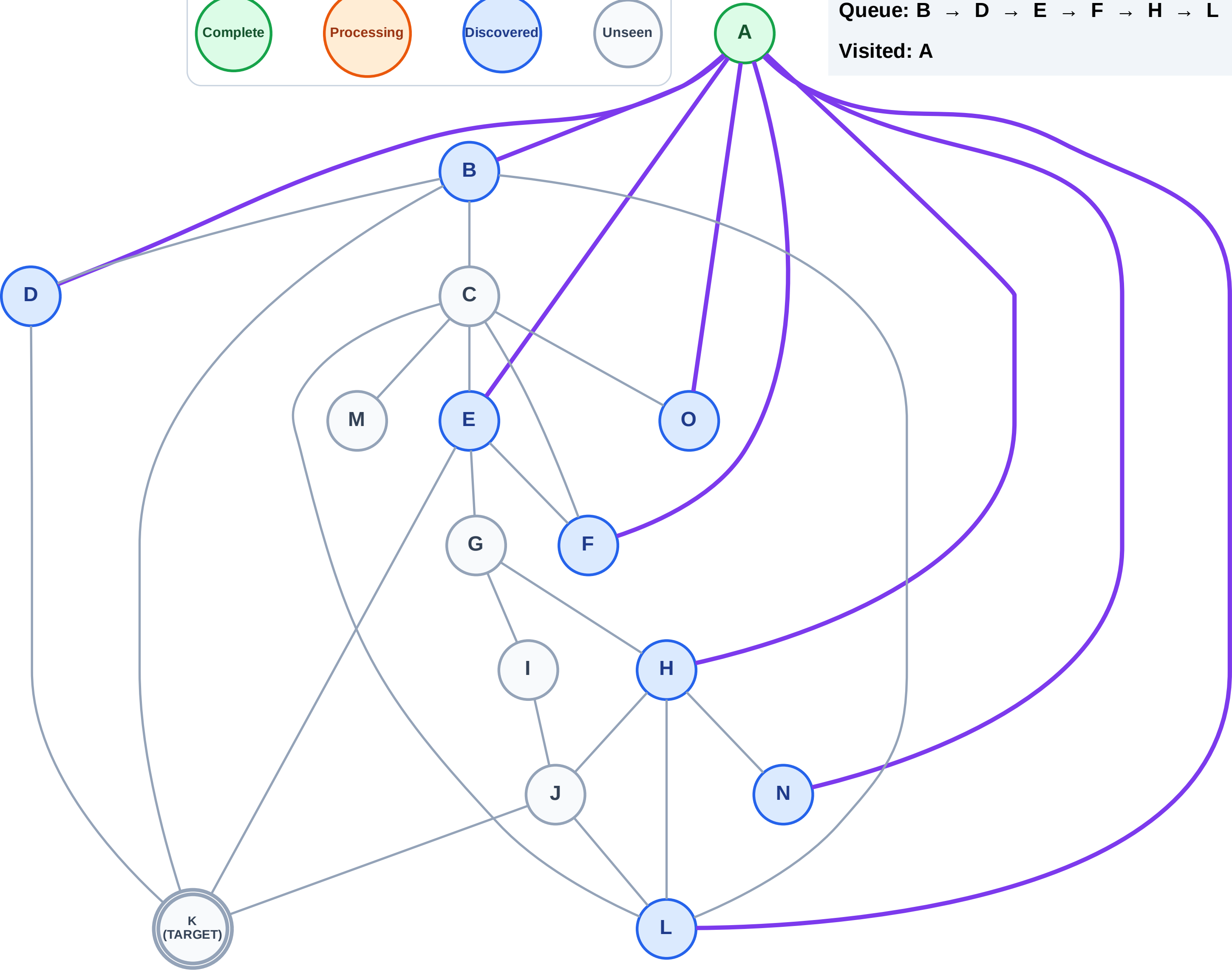
Processing

Discovered

Unseen

Queue: B → D → E → F → H → L → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

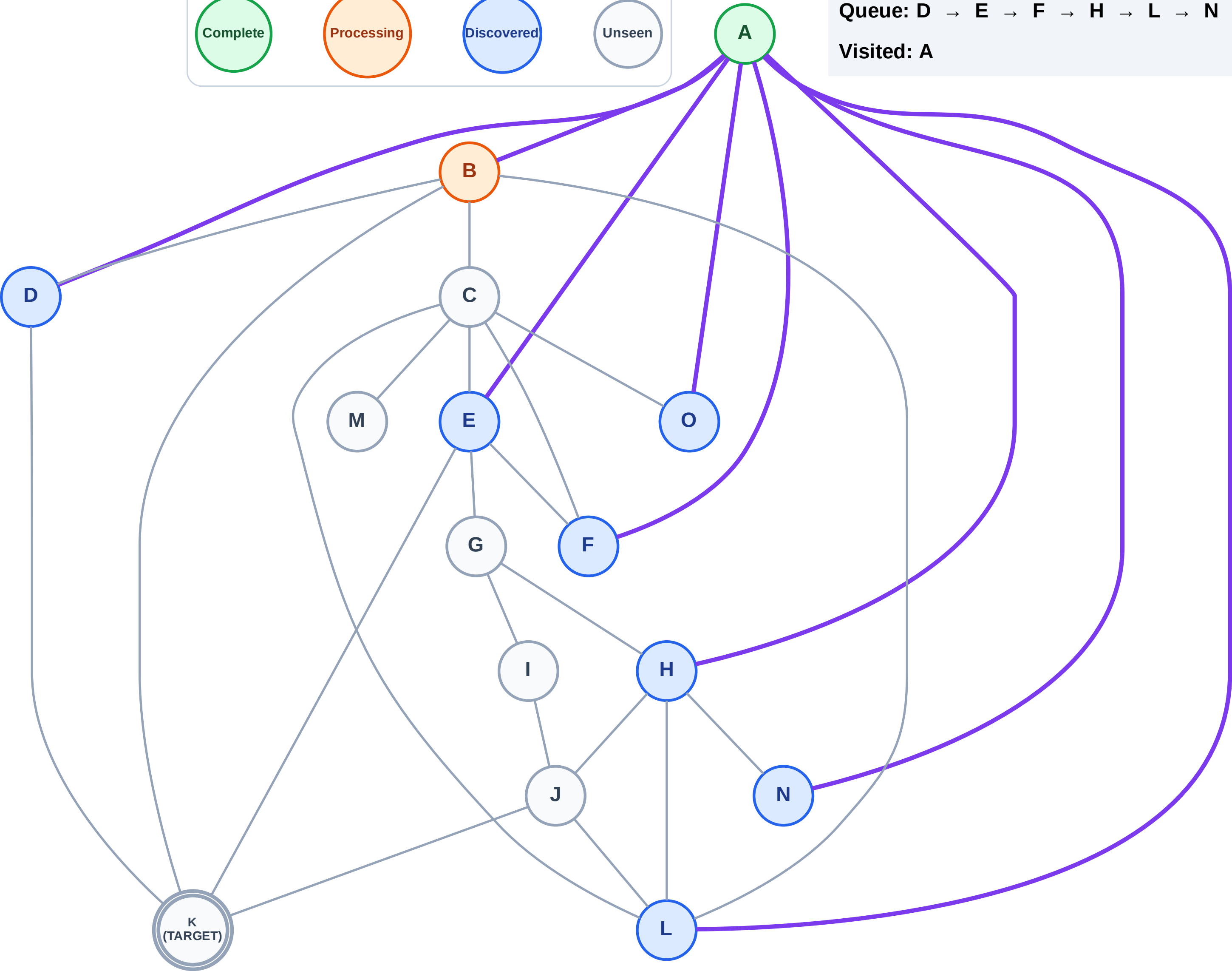
Processing

Discovered

Unseen

Queue: D → E → F → H → L → N → O

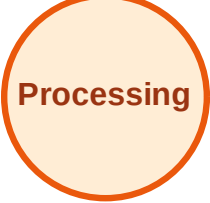
Visited: A



BFS Traversal

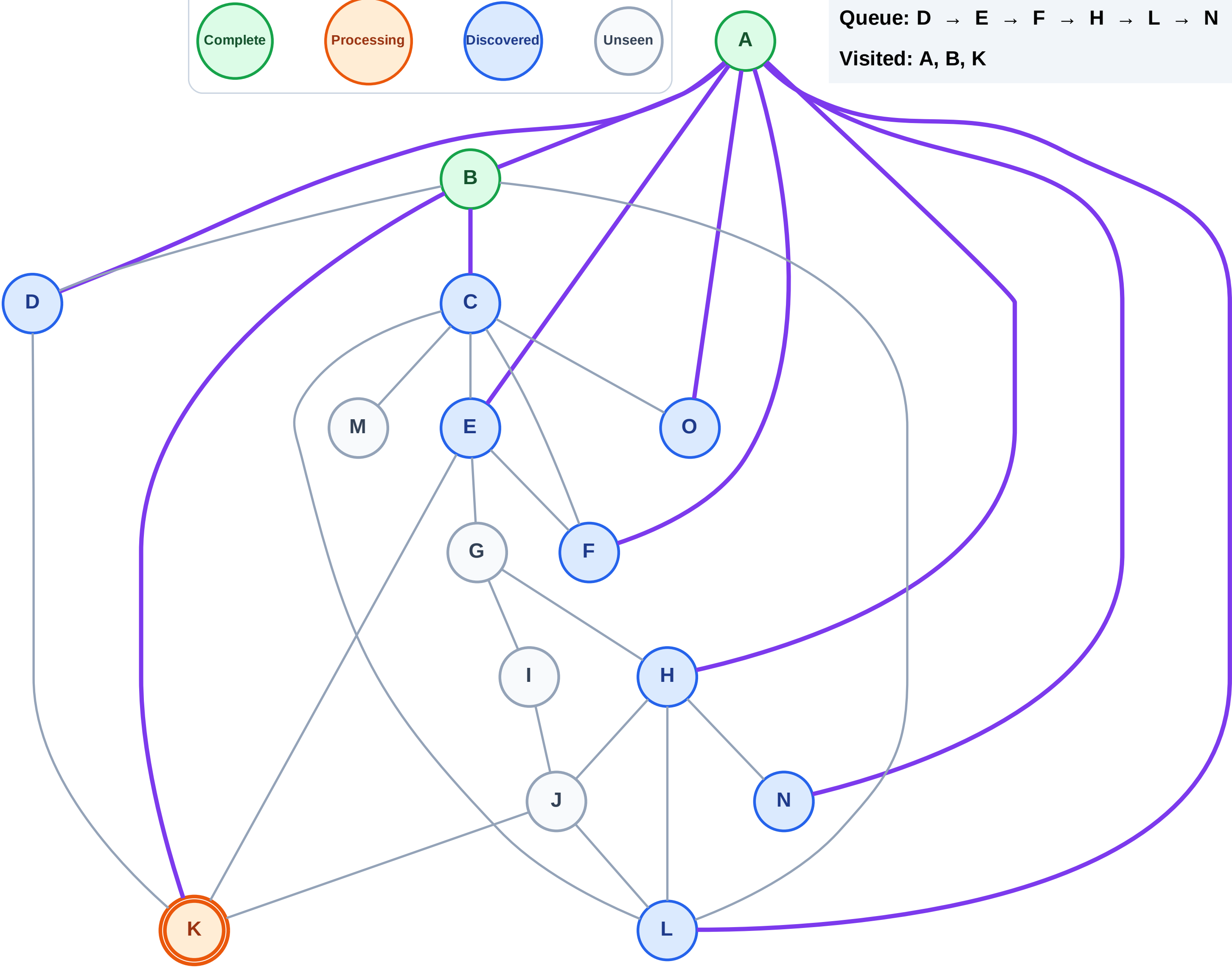
Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend



Queue: D → E → F → H → L → N → O → C

Visited: A, B, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

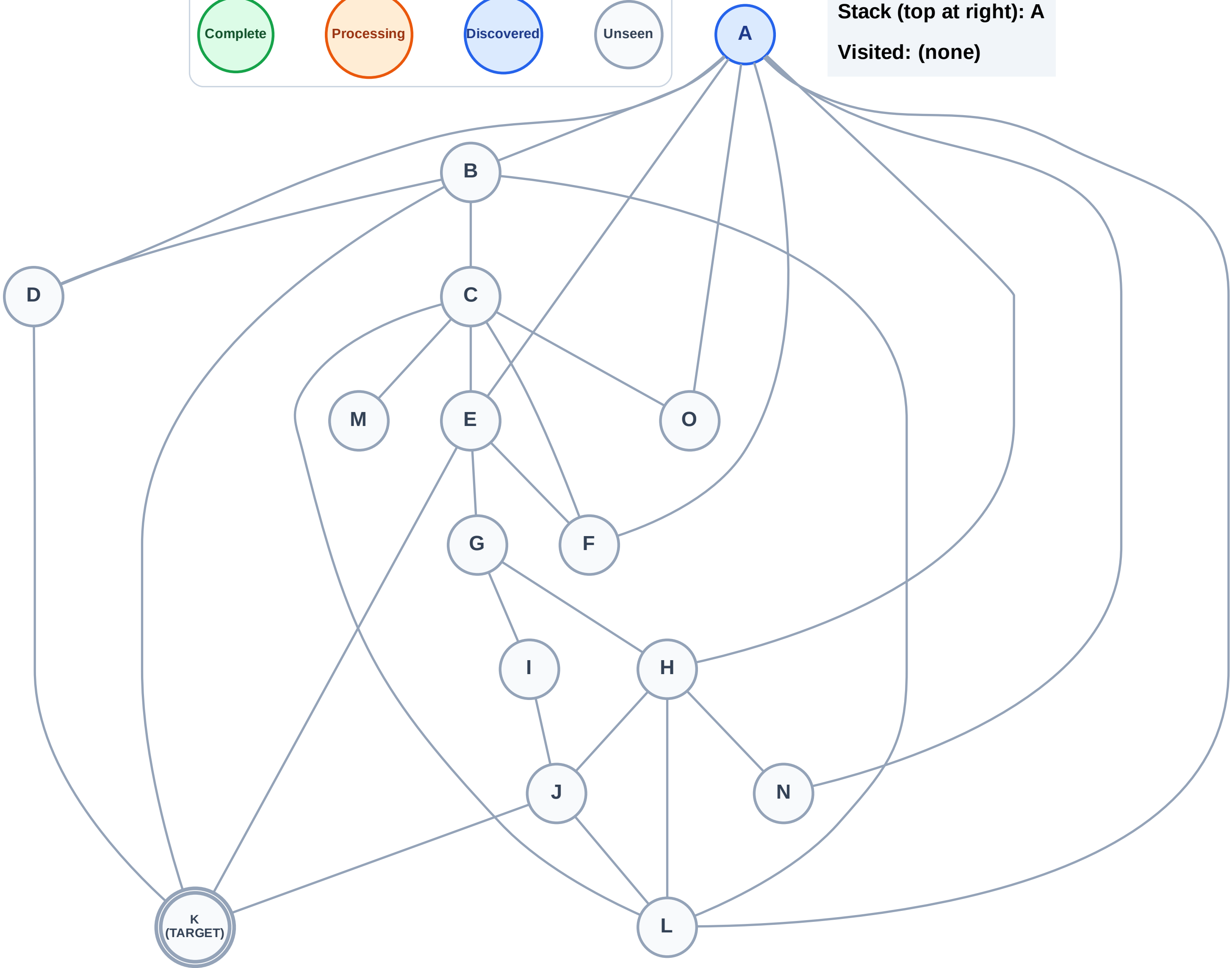
Complete

Processing

Discovered

Unseen

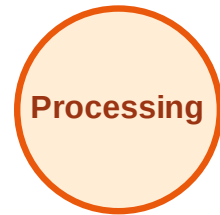
Stack (top at right): A
Visited: (none)



DFS Traversal

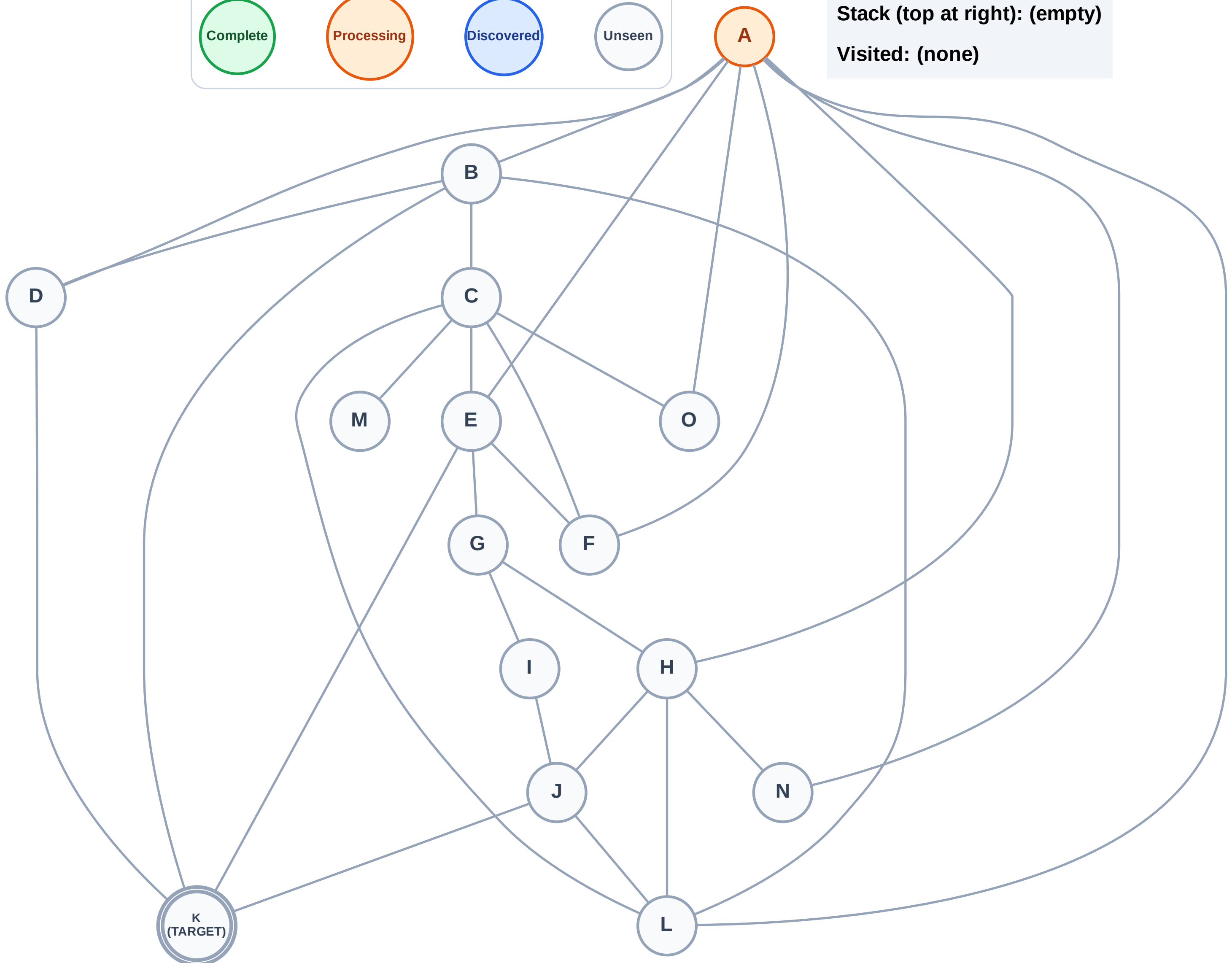
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover O, N, L, H, F, E, D, B from A

Legend

Complete

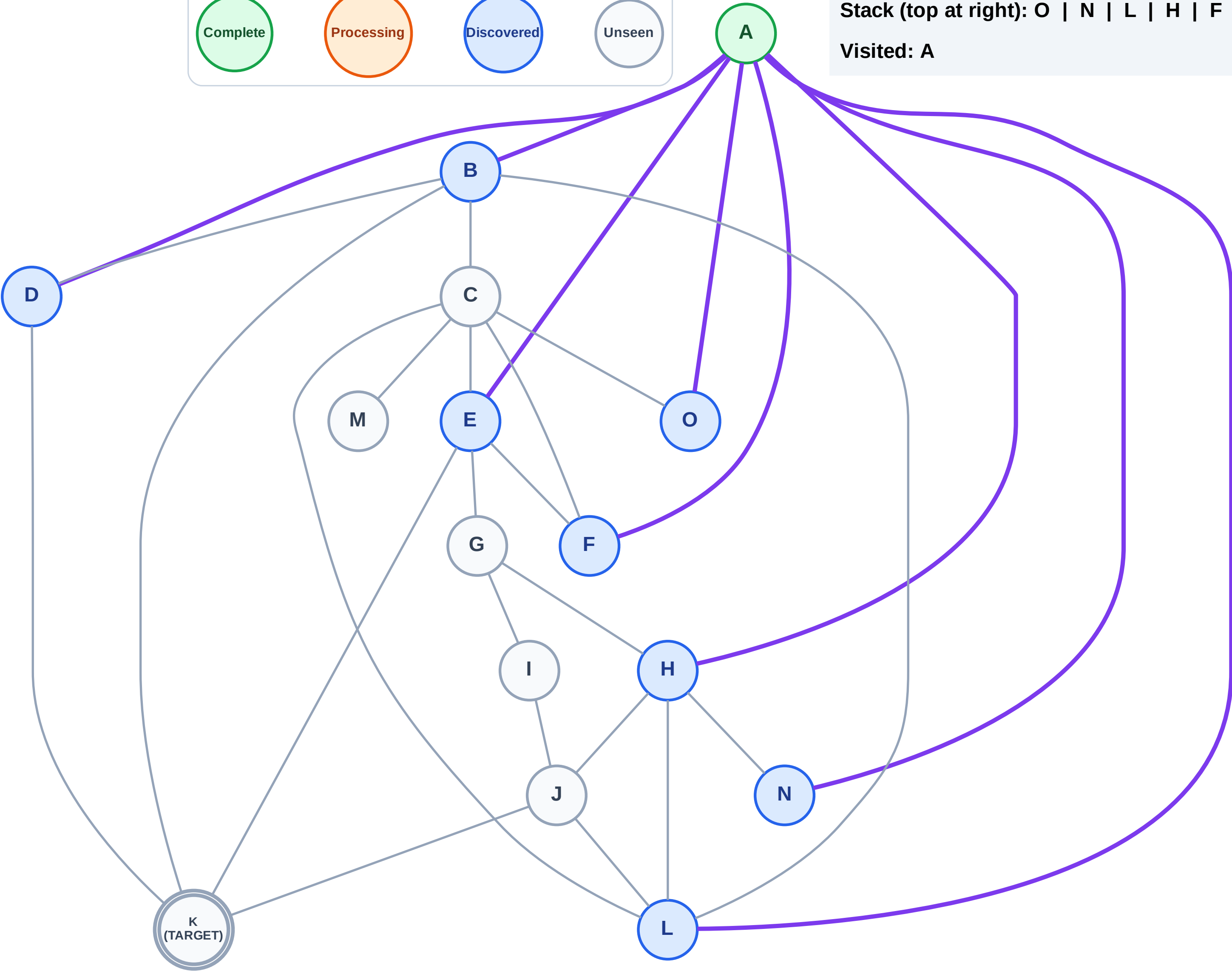
Processing

Discovered

Unseen

Stack (top at right): O | N | L | H | F | E | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

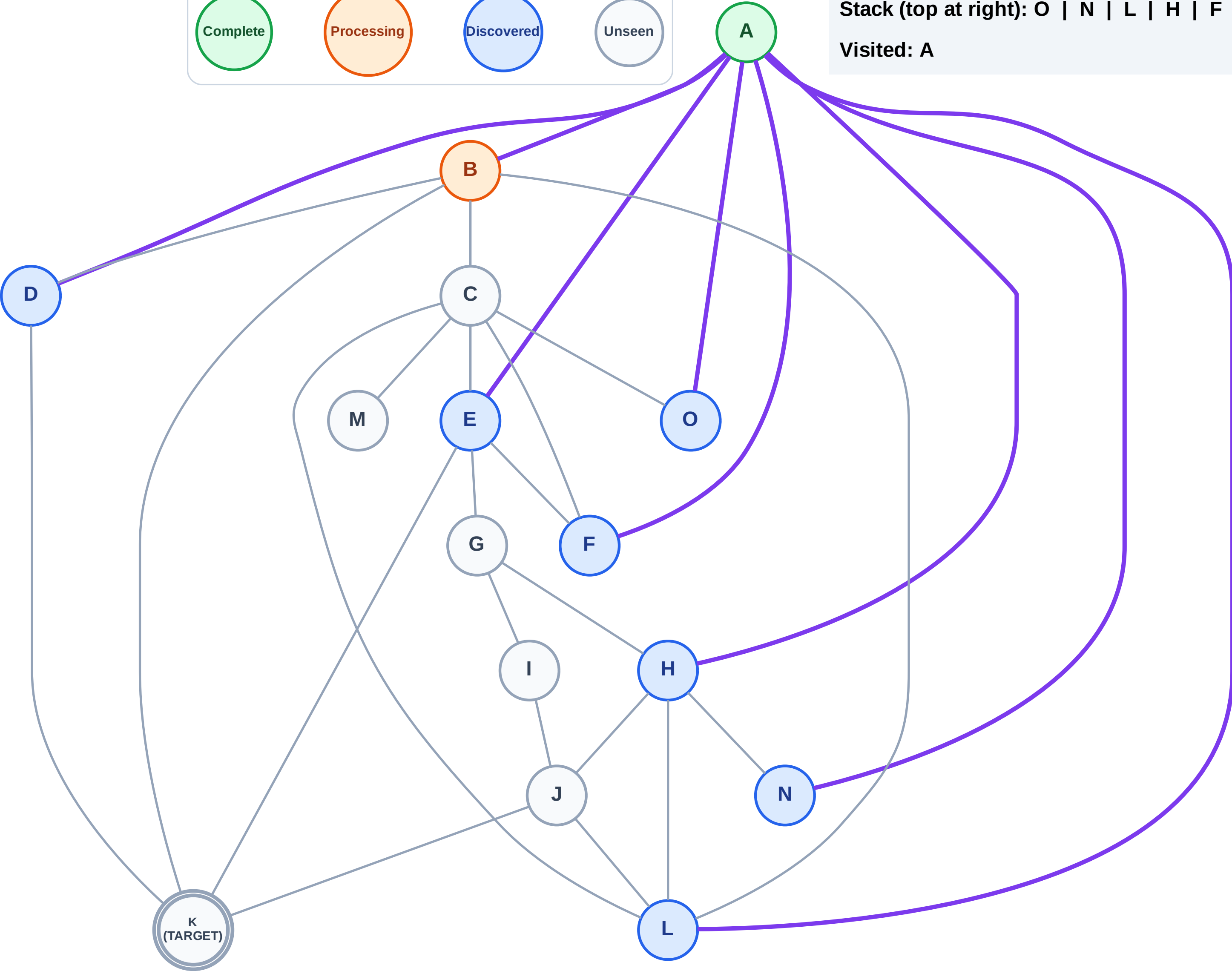
Processing

Discovered

Unseen

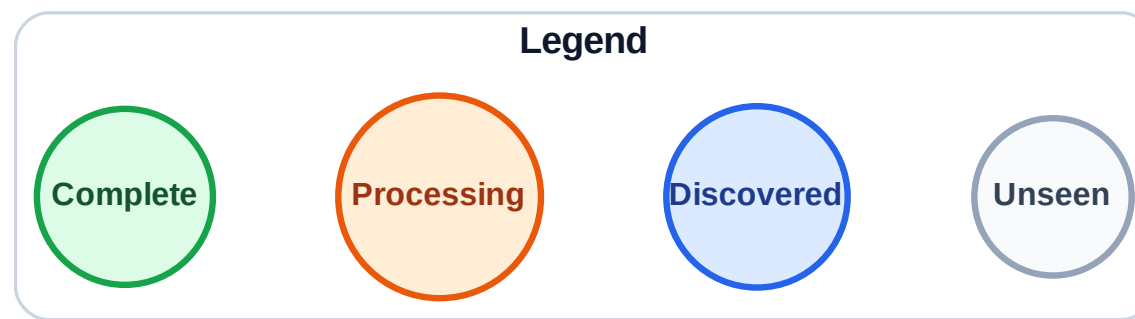
Stack (top at right): O | N | L | H | F | E | D

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K



Stack (top at right): O | N | L | H | F | E | D

Visited: A, B, K

